

Mark R. Saddler

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EDUCATION

Massachusetts Institute of Technology Ph.D. in Computational Neuroscience Thesis: "Task-optimized models of human hearing link perception and neural coding"	Cambridge, MA 02/2024
The University of Chicago B.A. in Biological Sciences (specialization in neuroscience) B.S. in Biological Chemistry B.A. in Chemistry	Chicago, IL 06/2017

RESEARCH POSITIONS

DTU Department of Health Technology , Hearing Systems Research Section <i>Postdoc</i> (advised by Torsten Dau)	Lyngby, Denmark 02/2024 – Present
MIT Department of Brain and Cognitive Sciences , Lab for Computational Audition <i>Ph.D. Student</i> (advised by Josh H. McDermott)	Cambridge, MA 10/2017 – 02/2024
Meta Reality Labs , Audio Research <i>Research Scientist Intern</i> (advised by Antje Ihlefeld)	Redmond, WA 05/2022 - 09/2022
Woods Hole Oceanographic Institution , Biology Department <i>Summer Student Fellow</i> (advised by Laela Sayigh)	Woods Hole, MA 06/2016 - 08/2016
UChicago Department of Neurobiology , Epilepsy Lab <i>Undergraduate Research Assistant</i> (advised by Wim van Drongelen)	Chicago, IL 10/2015 - 06/2017
UCSD Scripps Institution of Oceanography , Marine Physical Laboratory <i>Summer Research Intern</i> (advised by Simone Baumann-Pickering)	San Diego, CA 06/2015 - 09/2015
Salk Institute for Biological Studies , Molecular Neurobiology Laboratory <i>Summer Research Intern</i> (advised by Laura Hale and Sreekanth Chalasani)	San Diego, CA 06/2014 - 09/2014

PUBLICATIONS

4. **MR Saddler**, JH McDermott (2024). Models optimized for real-world tasks reveal the task-dependent necessity of precise temporal coding in hearing. *Nature Communications* 15, 10590.
3. **MR Saddler***, R Gonzalez*, JH McDermott (2021). Deep neural network models reveal interplay of peripheral coding and stimulus statistics in pitch perception. *Nature Communications* 12, 7278.
2. **MR Saddler***, A Franci*, J Feather, K Qian, Y Zhang, JH McDermott (2021). Speech denoising with auditory models. *Proc. Interspeech* 2021, 2681-2685.
1. **MR Saddler**, A Bocconcelli, L Hickmott, G Chiang, R Landea-Briones, P Bahamonde, G Howes, P Segre, L Sayigh (2017). Characterizing Chilean blue whale vocalizations with DTAGs: a test of using tag accelerometers for caller identification. *Journal of Experimental Biology* 220, 4119-4129.

PREPRINTS AND CONFERENCE PROCEEDINGS

6. **MR Saddler**, T Dau, JH McDermott (2025). Towards individualized models of hearing-impaired speech perception. *Proceedings of the 6th Clarity Workshop*.
5. A Banerjee, **MR Saddler**, JG Arenberg, JH McDermott (2025). A deep learning framework for understanding cochlear implants. *bioRxiv*.
4. **MR Saddler**, T Dau, JH McDermott (2024). Psychoacoustic phenomena explained by auditory task optimization. *Proceedings of the Conference on Cognitive Computational Neuroscience*.
3. **MR Saddler**, M Khaleghi-Meybodi, A Ihlefeld (2023). Automated audio realism detection. *Technical Disclosure Commons*.
2. J Suresh, **MR Saddler**, V Bindokas, A Bhansali, L Pesce, J Wang, J Marks, W van Drongelen (2023). Emerging activity patterns and synaptogenesis in dissociated hippocampal cultures. *bioRxiv*.
1. A Bocconcelli, L Hickmott, G Chiang, P Bahamonde, G Howes, R Landea-Briones, F Caruso, **MR Saddler**, L Sayigh (2016). DTAG studies of blue whales (*Balaenoptera musculus*) in the Gulf of Corcovado, Chile. *Proceedings of Meetings on Acoustics* 27, 040002.

FUNDING AND AWARDS

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|------|---|
| 2025 | William Demant Foundation postdoctoral research grant (2.9 million DKK ~ \$450,000 over 3 years) |
| 2023 | Travel Award (Association for Research in Otolaryngology, Orlando) |
| 2022 | Best Video Pitch Award (Virtual Conference on Computational Audiology) |
| 2022 | Travel Award (Association for Research in Otolaryngology) |
| 2021 | K. Lisa Yang Integrative Computational Neuroscience (ICoN) Center Graduate Student Fellowship |
| 2021 | Travel Grant (Interspeech 2021, Brno, Czechia) |
| 2020 | Best Poster Award (Advances and Perspectives in Auditory Neuroscience) |
| 2020 | Angus MacDonald Award for Excellence in Undergraduate Teaching (MIT) |
| 2019 | Best Poster Award (IBM AI Week, Cambridge) |
| 2019 | Travel Award (Association for Research in Otolaryngology, Baltimore) |
| 2017 | Best Student Paper Award (Acoustical Society of America, Boston) |
| 2017 | Phi Beta Kappa (The University of Chicago) |
| 2017 | Francis E. Knock Prize in Biological Chemistry for highest GPA in major (The University of Chicago) |
| 2016 | Woods Hole Oceanographic Institution Academic Programs Office Conference Travel Grant |
| 2016 | Woods Hole Oceanographic Institution Summer Student Fellowship |
| 2013 | University Scholar Award (The University of Chicago) |
| 2013 | National Merit Scholarship |

INVITED TALKS

10. ARO Symposium, *Pitch: New Perspectives on a Classic Problem*. San Juan, Puerto Rico (February, 2026)
9. Foundation Pour L'Audition, *Auditory Horizons Workshop on AI*. Paris, France (November, 2025)
8. Institut De L'Audition, *IDA-DTU-ENS Workshop*. Paris, France (November, 2024)
7. DTU, *Hearing Systems Presentation Day*. Lyngby, Denmark (November, 2024)
6. WSAudiology. Lynge, Denmark (October, 2024)
5. Universität Oldenburg, *Dept. of Medical and Physical Acoustics Symposium*. Oldenburg, Germany (July, 2024)
4. DTU, *Hearing Systems Research Seminar*. Lyngby, Denmark (June, 2023)
3. Universität Graz, *Research Seminar in Systematic Musicology*. Graz, Austria (June, 2023)

2. Ecole Normale Supérieure, *Laboratoire des Systèmes Perceptifs*. Paris, France (April, 2022)
1. MIT Department of Brain and Cognitive Sciences, *Cog Lunch*. Cambridge, MA (April, 2022)

TEACHING

DTU Department of Health Technology

- *Guest Lecture on Machine Learning: Auditory Signal Processing and Perception* (22003) 05/2025

MIT Department of Brain and Cognitive Sciences

- TA: Machine-Motivated Human Vision (9.60) Spring 2023
- TA: Machine-Motivated Human Vision (9.60) Spring 2022
- TA: Introduction to Neural Computation (9.40) Spring 2020
- TA: Perception (9.35) Spring 2019

MIT Educational Studies Program

- *HSSP Teacher: Sensational Neuroscience* 06/2019 - 08/2019
- *Splash! Instructor: Braaaaaains! The Neuroscience of Zombies* 11/2018
- *HSSP Teacher: The Foundations of Intelligence: An Intro to Neuroscience* 06/2018 - 08/2018
- *HSSP Teacher: Memory and Mind Control: An Intro to Neuroscience* 02/2018 - 04/2018
- *Splash! Instructor: Introduction to Neuroscience* 11/2017

UChicago Biological Sciences Collegiate Division

- *Lab TA: Biological Dynamics (BIOS 20236)* Spring 2017
- *Lab TA: Principles of Physiology (BIOS 20242)* Fall 2016
- *Lecture TA: Principles of Physiology (BIOS 20242)* Fall 2015

MENTORING AND SERVICE

Journal reviewing (ad hoc) 06/2019 – Present
Journal of the Acoustical Society of America; Nature Communications; Nature Machine Intelligence; eLife; Attention, Perception, & Psychophysics; IEEE Transactions on Audio, Speech and Language Processing; Trends in Hearing; PLoS Computational Biology

Conference organizing

ISAAR session chair for “Computational Audiology, AI, and Auditory Profiling” 08/2025

MIT OpenMind High-Performance Computing Cluster

Group Representative Cambridge, MA
 06/2019 – 02/2024

MIT Division of Student Life

Graduate Resident Advisor in Baker House Cambridge, MA
 08/2018 - 07/2023

MIT Center for Brains Minds + Machines

Summer Research Program Mentor Cambridge, MA
 04/2020 - 09/2021

UChicago College Housing

Assistant Resident Head (RA) of Thangaraj / Tufts House Chicago, IL
 04/2015 - 06/2017

UChicago Emergency Medical Service

Emergency Medical Responder (EMR) Chicago, IL
 03/2014 - 01/2017

CONFERENCE PRESENTATIONS

40. **MR Saddler**, T Dau, JH McDermott (August, 2025). Towards individualized models of hearing-impaired speech perception. *The 6th Clarity Workshop on Improving Speech-in-Noise for Hearing Devices*. Delft, The Netherlands. **(Talk)**
39. **MR Saddler**, T Dau, JH McDermott (August, 2025). Modeling normal and impaired hearing with deep neural networks optimized for ecological tasks. *International Symposium on Auditory and Audiological Research*. Nyborg, Denmark. **(Poster)**
38. **MR Saddler**, T Dau, JH McDermott (June, 2025). Modeling normal and impaired hearing with deep neural networks optimized for ecological tasks. *International Symposium on Hearing*. Vienna, Austria. **(Talk)**
37. **MR Saddler**, T Dau, JH McDermott (March, 2025). Modeling normal and impaired hearing with deep neural networks optimized for ecological tasks. *DAS-DAGA Joint Annual Meeting of the German and Danish Acoustical Societies*. Copenhagen, Denmark. **(Talk)**
36. **MR Saddler**, T Dau, JH McDermott (February, 2025). Modeling normal and impaired hearing with deep neural networks optimized for ecological tasks. *Association for Research in Otolaryngology*. Orlando, FL. **(Poster blitz)**
35. A Banerjee, **MR Saddler**, JG Arenberg, JH McDermott (February, 2025). Understanding cochlear implants using machine learning. *Association for Research in Otolaryngology*. Orlando, FL. **(Poster)**
34. A Magaro, EN Shook, AJE Kell, **MR Saddler**, G Elbanna, JH McDermott (February, 2025). Optimization under ecological realism reproduces signatures of human speech perception. *Association for Research in Otolaryngology*. Anaheim, CA. **(Poster)**
33. **MR Saddler**, T Dau, JH McDermott (August, 2024). Psychoacoustic phenomena explained by auditory task optimization. *Cognitive Computational Neuroscience*. Boston, MA. **(Poster)**
32. A Banerjee, **MR Saddler**, JH McDermott (August, 2024). Neural network models of hearing clarify factors limiting cochlear implant outcomes. *Cognitive Computational Neuroscience*. Boston, MA. **(Poster)**
31. **MR Saddler**, T Dau, JH McDermott (June, 2024). Modeling normal and impaired hearing with artificial neural networks optimized for ecological tasks. *Virtual Conference on Computational Audiology*. **(Talk)**
30. A Banerjee, **MR Saddler**, JH McDermott (June, 2024). Neural network models of hearing clarify factors limiting cochlear implant outcomes. *Virtual Conference on Computational Audiology*. **(Talk)**
29. **MR Saddler**, JH McDermott (February, 2024). The role of temporal coding in hearing: evidence from task optimization. *Association for Research in Otolaryngology*. Anaheim, CA. **(Poster)**
28. A Banerjee, **MR Saddler**, JH McDermott (February, 2024). Understanding cochlear implants using machine learning. *Association for Research in Otolaryngology*. Anaheim, CA. **(Poster)**
27. A Magaro, EN Shook, AJE Kell, **MR Saddler**, JH McDermott (February, 2024). Optimization under ecological realism reproduces signatures of human speech perception. *Association for Research in Otolaryngology*. Anaheim, CA. **(Poster)**
26. **MR Saddler**, JH McDermott (August, 2023). The role of temporal coding in real-world hearing: evidence from machine learning. *International Symposium on Auditory and Audiological Research*. Nyborg, Denmark. **(Talk)**
25. A Banerjee, **MR Saddler**, JH McDermott (July, 2023). Neural network models clarify the role of plasticity in cochlear implant outcomes. *Conference on Implantable Auditory Prostheses*. Lake Tahoe, CA. **(Poster)**

24. **MR Saddler**, JH McDermott (February, 2023). The role of temporal coding in real-world hearing: evidence from machine learning. *Association for Research in Otolaryngology*. Orlando, FL. **(Poster)**
23. A Banerjee, **MR Saddler**, JH McDermott (February, 2023). Neural network models of hearing through a cochlear implant. *Association for Research in Otolaryngology*. Orlando, FL. **(Poster)**
22. Griffith IM*, McPherson MJ*, **MR Saddler***, JH McDermott (February, 2023). Task-optimized models of relative pitch. *Association for Research in Otolaryngology*. Orlando, FL. **(Poster)**
21. **MR Saddler**, JH McDermott (November, 2022). The role of temporal coding in real-world hearing: evidence from machine learning. *Advances and Perspectives in Auditory Neuroscience*. San Diego, CA. **(Poster)**
20. **MR Saddler**, JH McDermott (July, 2022). The role of temporal coding in everyday hearing: evidence from machine learning. *Virtual Conference on Computational Audiology*. **(Talk)**
19. **MR Saddler**, JH McDermott (March, 2022). The role of temporal coding in everyday hearing: evidence from deep neural networks. *Computational and Systems Neuroscience (COSYNE)*. Lisbon, Portugal. **(Poster)**
18. **MR Saddler**, JH McDermott (February, 2022). The role of auditory nerve phase locking in audition: evidence from deep neural networks. *Association for Research in Otolaryngology*. **(Poster)**
17. **MR Saddler**, JH McDermott (November, 2021). The role of auditory nerve phase-locking in human hearing: evidence from deep neural networks. *Advances and Perspectives in Auditory Neuroscience*. **(Talk)**
16. **MR Saddler***, A Franci*, J Feather, K Qian, Y Zhang, JH McDermott (August, 2021). Speech denoising with auditory models. *Interspeech*. Brno, Czechia. **(Poster)**
15. **MR Saddler**, J Feather, A Franci, JH McDermott (June, 2021). Hearing-impaired artificial neural networks replicate speech recognition deficits of hearing-impaired humans. *Virtual Conference on Computational Audiology*. **(Talk)**
14. **MR Saddler**, R Gonzalez, JH McDermott (February, 2021). Deep neural networks reveal interplay of peripheral coding and stimulus statistics in human pitch perception. *Association for Research in Otolaryngology*. **(Poster)**
13. **MR Saddler**, JH McDermott (February, 2021). The role of auditory nerve phase-locking in human hearing: evidence from deep neural networks. *Association for Research in Otolaryngology*. **(Poster)**
12. **MR Saddler**, R Gonzalez, JH McDermott (October, 2020). Deep neural networks reveal interplay of peripheral coding and stimulus statistics in human pitch perception. *Advances and Perspectives in Auditory Neuroscience*. **(Poster)**
11. Medina B, **MR Saddler**, JH McDermott (October, 2020). Investigating deep artificial neural networks trained to do ecological tasks as a normative model for pitch perception. *Society for Advancement of Chicanos/Hispanics and Native Americans in Science*. **(Poster)**
10. Medina B, **MR Saddler**, JH McDermott (October, 2020). Investigating deep artificial neural networks trained to do ecological tasks as a normative model for pitch perception. *Central European Conference on Information and Intelligent Systems*. **(Poster)**
9. **MR Saddler**, J Feather, A Franci, JH McDermott (March, 2020). Hearing-impaired deep neural networks replicate behavioral deficits of hearing-impaired humans. *Computational and Systems Neuroscience (COSYNE)*. Denver, CO. **(Poster)**
8. **MR Saddler**, J Feather, A Franci, R Gonzalez, JH McDermott (January, 2020). Deep neural networks with simulated hearing impairment replicate behavioral deficits of hearing-impaired listeners. *Association for Research in Otolaryngology*. San Jose, CA. **(Talk)**

7. **MR Saddler**, R Gonzalez, JH McDermott (October, 2019). Characteristics of human pitch perception emerge in neural networks optimized to estimate F0 from natural sounds. *Advances and Perspectives in Auditory Neuroscience*. Chicago, IL. **(Poster)**
6. JH McDermott, J Feather, A Franci, **MR Saddler**, Y Zhang, H Sarker (September, 2019). Next-generation hearing aids via neural network models of the auditory system. *MIT-IBM AI Horizons Colloquium*. Cambridge, MA. **(Poster)**
5. **MR Saddler**, R Gonzalez, JH McDermott (February, 2019). Neural networks trained to estimate F0 from natural sounds replicate properties of human pitch perception. *Association for Research in Otolaryngology*. Baltimore, MD. **(Poster)**
4. **MR Saddler**, R Gonzalez, JH McDermott (November, 2018). neural networks trained to estimate F0 from natural sounds replicate properties of human pitch perception. *MIT Quest Symposium on Robust, Interpretable Deep Learning Systems*. Cambridge, MA. **(Poster)**
3. **MR Saddler**, R Gonzalez, JH McDermott (October, 2018). Neural networks trained to estimate f0 from natural sounds replicate properties of human pitch perception. *Speech and Audio in the Northeast*. Cambridge, MA. **(Poster)**
2. **MR Saddler**, A Bocconcelli, L HickmottS, G Chiang, Landea-Briones R, P BahamondeA, Howes G, L Sayigh (June, 2017). Characterizing Chilean blue whale vocalizations with digital acoustic recording tags: a test of using tag accelerometers for caller identification. *Acoustical Society of America and European Acoustics Association*. Boston, MA. **(Poster)**
1. A Bocconcelli, **MR Saddler**, L Hickmott, G Chiang, P Bahamonde, F Caruso, L Sayigh (July, 2016). Acoustic behavior of blue whales (*Balaenoptera musculus*) in the Gulf of Corcovado, Chile, recorded on DTAGs. *4th International Conference on The Effects of Noise on Aquatic Life*. Dublin, Ireland. **(Poster)**

GRADUATE COURSEWORK

Cumulative GPA: 5.0 / 5.0

- Systems Neuroscience (9.011)
- Quantitative Methods and Computational Models in Neurosciences (9.014)
- Computational Cognitive Science (9.660)
- Applied Machine Learning (6.862)
- Audition: Neural Mechanisms, Perception and Cognition (9.285)
- Matrix Methods in Data Analysis, Signal Processing, and Machine Learning (18.0651)
- Automatic Speech Recognition (6.345)

SKILLS AND INTERESTS

Software: Python, MATLAB, Tensorflow, PyTorch, JAX, SLURM, online psychophysics (MTurk/Prolific), LaTeX

Interests: hiking, running, soccer, skiing, teaching, marine mammals